Fares Mohamed

Alexandria, Egypt | faresmohamed260@gmail.com | github.com/faresmohamed260
linkedin.com/in/fares-mohamed-b83454194 | **Robotics & Embedded AI Engineer**

Summary

Computer Science graduate ranked first in class, building embedded prototypes, robotics control interfaces, computer-vision systems, and production-minded AI software with Python, C++, ESP32, and modern web tooling.

Experience

Armstrong 2023–2025
AI & Automation Engineer / Technical Content Creator (Part-time)

- Built AI-assisted educational pipelines and developed robotics, Arduino, circuits, AI, and programming curricula with classroom-ready demonstrations and technical media.

Kayfa Aug–Dec 2025
AI Subject Matter Expert

- Authored practical ML lessons, assignments, and Python notebooks on neural networks, CNNs, clustering, and anomaly detection.

Engineering for Kids-Egypt 2023–2025
AI & Robotics Instructor (Seasonal)

- Taught students aged 11–16 robotics, Arduino, circuits, AI, Python, and C++; guided builds through assembly, integration, programming, debugging, and demonstrations.

Selected Projects

DUM-E – ESP32 Robotic Arm Platform
Python, C++, ESP32, Streamlit, Robotics

- Built firmware and an operator dashboard for calibration, manual control, guarded execution, controller mapping, sequence recording, and FK/IK-assisted pick-and-place.

S.A.G.A. – Narrative Intelligence Platform
Python, NLP, FastAPI, Durable Workers

- Designed outbound-only local analysis workers, durable cloud job control, typed evidence contracts, reproducible benchmarks, and selective local inference for long-form fiction.

RenderLab – AI Media Workspace
Next.js, TypeScript, ComfyUI, Modal, Supabase, R2

- Built browser-independent GPU job reconciliation, failover, media persistence, lifecycle controls, and operator/account safety for a production-domain closed beta.

Fares Uniform – Experimental ERP & Public Website
Odoo, Python, Next.js, PostgreSQL

- Engineered offline POS reconciliation, stock custody, roles, preorders, production handoff, B2B flows, WebSocket replay, cron locking, backup/restore, and bilingual browser validation.

VisionDeck – Computer Vision Suite
Python, OpenCV, MediaPipe, YOLO

- Developed modular real-time face, gesture, and object detection using webcam and Android IP-camera feeds.

Education

Alexandria University, Faculty of Computer and Data Science 2022–2026
B.Sc. Computer Science (Intelligent Systems) | **Class Rank: 1st**

Skills

Robotics/Embedded: Arduino, ESP32, C++, sensors, circuits, calibration, controller mapping, FK/IK-assisted motion
AI/Vision: Python, OpenCV, MediaPipe, YOLO, NLP, RAG, ComfyUI, ControlNet
Software/Platforms: FastAPI, Next.js/React, Streamlit, Docker, Git, pytest, PostgreSQL, Supabase, Modal | **Languages:** Arabic, English

Leadership & Certifications

Head of AI Committee, ElCoder – Led AI/ML workshops. | Machine Learning Specialization | Deep Learning Specialization | Google Cloud Big Data & ML Fundamentals